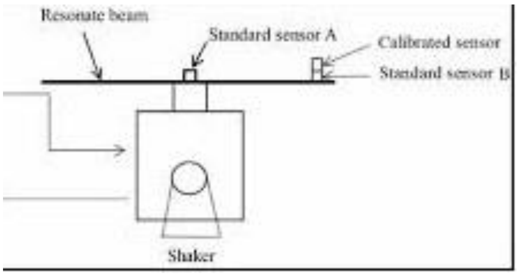
PROJECT ANNOUNCEMENT
Fall 2025

Title: DESIGN AND CONSTRUCTION OF AN ACCELEROMETER CALIBRATION SYSTEM



Position Ref: ACCALIB

Type: Individual
Status: New
Seats: 1 student

Project Description:
Inexpensive off-the-shelf MEMS accelerometers have large measurement noise and their calibration is crucial. On the other hand the hovercraft in the department needs accurate acceleration data to be properly controlled. The student working on the project will design and construct an accelerometer calibration platform. The project involves calibration of several inexpensive accelerometer units MPU6050 and deriving their statistical metrics. This project is related to another one offered this semester, which involves sensor fusion related to combined utilization of multiple sensors.

Work Description:

- Literature survey for relevant papers.
- Reviewing/learning metrics used for sensor characterization.
- Studying various methods of calibration.
- Designing a platform for accelerometer calibration.
- Construction of the platform.
- Running the platform at different acceleration values to characterize each accelerometer.
- Processing the data to derive statistical properties.

Required Qualifications and Skills:

- Ability and appetite to learn new subjects.
- Appetite for experimental and, to some extent, theoretical work.
- The project will be on campus.

Related UN Sustainable Development Goals:

- No.9 Industry, innovation and infrastructure.

Difficulty:	high ( )	medium (X)	low ( )
Time commitment:	high (X)	medium ( )	low ( )
Estimated expense:	0 TL		